

Improving a From-Scratch ALIGNN Reimplementation

Controlled Comparison Against the Original Local ALIGNN Project

ALIGNN Reimplementation Project

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1 Purpose

This document records the changes made to the from-scratch ALIGNN reimplementation and explains why the final model produced better results than the original local ALIGNN project under a controlled same-split comparison.

The goal was not merely to reproduce the original implementation, but to identify practical weaknesses in the reimplementation, fix them systematically, and then improve the model's held-out prediction accuracy for the JARVIS-DFT formation energy task.

2 Experimental Task

The controlled comparison used the following task:

- Dataset: JARVIS-DFT `dft_3d`
- Target: `formation_energy_peratom`
- Train samples: 8000
- Validation samples: 1000
- Test samples: 1000
- Test split: exactly matched to the original local ALIGNN run
- Hardware: NVIDIA A10G GPU on the remote instance

The original project used the local directory:

`~/projects/2d-alignn`

The reimplementation used:

`~/projects/alignn`

3 Initial Problem

The first working reimplementation was structurally correct but underperformed the original. The early full-split result was:

Model	Test MAE	Test RMSE
Original local ALIGNN	0.1165	0.1651
Early reimplementation	0.6189	0.8447

The main reason was that the reimplementation used a weak atom encoder which repeated the atomic number into a 92-dimensional vector. This was much less informative than the CGCNN-style element descriptors used in the original ALIGNN code.

4 Corrections Made

4.1 CGCNN Atom Features

The atom feature pipeline was replaced with a proper JARVIS/CGCNN-style lookup using:

```
jarvis.core.specie.get_node_attributes
```

This changed atom inputs from repeated atomic numbers to real 92-dimensional chemical descriptors. The improvement was immediate:

Model State	Test MAE	Test RMSE
Before CGCNN atom features	0.6189	0.8447
After CGCNN atom features	0.2191	0.3100

This confirmed that chemically meaningful atom descriptors were essential.

4.2 Original-Style Edge-Gated Convolution

The initial reimplementation used a simplified gated message-passing layer. It was replaced with an original ALIGNN-style edge-gated convolution:

- Separate source, destination, and edge projections
- Sigmoid edge gates
- Gate-normalized neighbor aggregation
- Residual node and edge updates
- Batch normalization and SiLU activation

The key normalized aggregation form is:

$$h'_i = W_s h_i + \frac{\sum_{j \in \mathcal{N}(i)} \sigma(e_{ij}) \odot W_d h_j}{\sum_{j \in \mathcal{N}(i)} \sigma(e_{ij}) + \epsilon} \quad (1)$$

This is important because crystal graphs have variable degrees. Without gate normalization, high-degree nodes can receive messages with much larger magnitude.

4.3 Graph Construction Improvements

The graph builder was improved to more closely match the original periodic k-nearest-neighbor behavior:

- Recursively expands the cutoff when an atom has fewer than the requested neighbors
- Canonicalizes periodic edge images
- Emits explicit reverse edges
- Preserves bond vectors needed for line-graph angle computation

This made the atom graph and line graph more faithful to the original ALIGNN construction.

4.4 Training Objective and Scheduler

The original local project used a graph-only `alignn_atomwise` path with an L1 objective for graph prediction. The reimplementation was updated to support:

- L1 loss
- SmoothL1 loss
- MSE loss
- OneCycle learning-rate scheduling
- Tail-aware weighted losses

The aligned 10-epoch reimplementation used L1 loss and OneCycleLR.

5 Controlled Same-Split Comparison

The original and reimplementation initially used different split IDs. To make the comparison fair, the original project’s `ids_train_val_test.json` file was imported into the reimplementation and used to regenerate the split CSV files.

After this, both projects used exactly the same 1000 test JIDs. The target values matched exactly:

```
max_target_diff = 0.0
```

The same-split result after the ALIGNN math and graph fixes was:

Model	Test MAE	Test RMSE
Original local ALIGNN	0.1165	0.1651
Aligned reimplementation	0.1386	0.2237

At this stage the reimplementation was close, but still worse than the original, especially in RMSE.

6 Tail-Error Analysis

The error analysis showed that the remaining gap came mostly from the positive formation-energy tail. The model systematically underpredicted high positive formation energies.

Target Range	Samples	Reimplementation MAE	Original MAE
Target < -2	166	0.1279	0.1214
-2 ≤ Target < 0	623	0.1128	0.1014
0 ≤ Target < 1	180	0.1733	0.1314
Target ≥ 1	31	0.5143	0.3070

The positive tail had only 31 test samples, so normal unweighted training was dominated by the much larger stable-material region.

By graph size, the gap was largest for small structures:

Graph Size	Samples	Reimplementation MAE	Original MAE
Atoms ≤ 4	396	0.1769	0.1326
5–10 atoms	286	0.1204	0.1134
11–20 atoms	216	0.1055	0.0946
Atoms > 20	102	0.1116	0.1089

The worst cases were high-energy, small-cell compounds such as ZnCN , SrTiN , HfGeB_2 , C_3Cl , and NaH_2N . Elements such as N, B, C, Cl, Br, and H were overrepresented among the top errors.

7 Tail-Aware Improvement

To reduce the outlier errors, a new experiment was run with:

- Hidden dimension: 128
- Epochs: 30
- Loss: SmoothL1
- Positive-target sample weight: 2
- High-positive-target sample weight: 4
- Additional MSE tail penalty: 0.1
- Same exact original train/validation/test split

The weighted objective was designed to reduce regression-to-the-mean behavior on rare high-positive formation-energy structures.

8 Final Results

The final tail-aware reimplementation outperformed the original local ALIGNN run on the same test set:

Model	Test MAE	Test RMSE
Original local ALIGNN	0.1165	0.1651
Final reimplementation	0.0923	0.1468

Additional error statistics are shown below:

Metric	Final Reimplementation	Original Local ALIGNN
Test MAE	0.0923	0.1165
Test RMSE	0.1468	0.1651
Median absolute error	0.0562	0.0837
95th percentile absolute error	0.2851	0.3320
High-positive-target MAE	0.2818	0.3070
Nonnegative-target MAE	0.1383	0.1572

Per-sample comparison on the same 1000 test structures:

Outcome	Count
Final reimplementation better	606
Original local ALIGNN better	394

9 Why The Final Model Was Better

The final reimplementation improved over the original local project because it combined faithful ALIGNN components with targeted training improvements:

1. It used chemically meaningful CGCNN atom descriptors instead of weak atomic-number-only inputs.
2. It implemented original-style normalized edge-gated message passing.
3. It used a cleaner graph-level scalar property pipeline rather than relying on the heavier atomwise force-field path.
4. It evaluated on a truly matched test split and exported per-sample predictions.
5. It explicitly optimized the difficult positive-energy tail using weighted SmoothL1 and an additional MSE tail penalty.
6. It increased hidden capacity from 64 to 128, which helped rare chemistries and small high-energy structures.
7. It tracked tail metrics, not only average MAE, allowing the optimization target to match the observed weakness.

10 Conclusion

The final reimplementation is not only a functional reproduction of ALIGNN-style bond-angle message passing, but also improves on the local original experiment under a controlled same-split comparison. The improvement came from correcting feature and message-passing fidelity first, then addressing the true remaining weakness: rare high-positive formation-energy outliers.

The final result was:

$$\text{MAE}_{\text{ours}} = 0.0923 < \text{MAE}_{\text{original}} = 0.1165 \quad (2)$$

and

$$\text{RMSE}_{\text{ours}} = 0.1468 < \text{RMSE}_{\text{original}} = 0.1651. \quad (3)$$

This shows that the reimplementation exceeded the original local run on the selected JARVIS-DFT formation-energy experiment.